

Keywords: time series foundation models • zero-shot forecasting • benchmarking • temporal inductive bias

Foundations without Fundamentals: Zero-Shot Blind Spots in Time Series Foundation Models

Chronos-2, Moirai, and Toto Fail Unit Tests for Monotonic Trends, Periodic Signals, and Leading Indicators That Classical Models Pass Trivially

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arXiv:2610.02058 • October 2026

Despite strong performance on broad benchmarks, prominent Time Series Foundation Models fail surprisingly basic unit tests—monotonic trends, periodic signals, and leading-indicator exploitation—that simple classical models handle trivially, revealing a gap between pre-training scale and fundamental temporal inductive biases that persists on real-world sensor data.

AT A GLANCE

Problem: TSFMs may lack inductive biases for basic temporal patterns despite strong broad-benchmark performance.

Why it matters: Monotonic trends, periodicity, and causal leading indicators are ubiquitous in industrial, scientific, and healthcare time series.

Method: SimpleTimeBench—a diagnostic suite of unit tests for temporal primitives (trend, periodicity, leading indicators) at multiple difficulty levels.

Result: Chronos-2, Moirai, and Toto frequently fail zero-shot unit tests; ARIMA/Theta match or exceed them on basic primitives.

Evidence: Synthetic controlled series plus real-world building energy meters and meteorological stations.

The Big Picture

Time Series Foundation Models (TSFMs) trained on massive corpora—Chronos-2 on billions of time points, Moirai on diverse multi-domain data—have achieved impressive zero-shot performance on established benchmarks such as Monash and ETT. The natural inference is that these models have internalized general temporal structure, including basic primitives like monotonic trends, cyclical seasonality, and causal leading indicators.

This paper challenges that inference by constructing Sim-

pleTimeBench: a diagnostic benchmark of unit tests for temporal primitives where any well-calibrated forecaster should produce near-perfect zero-shot results. If a model with billions of training examples fails these trivial tests, the gap between pre-training scale and genuine temporal reasoning is larger than aggregate benchmarks suggest.

Why Existing Approaches Fall Short

Conflation problem: Aggregate benchmark performance combines hundreds of heterogeneous series; strong performance on diverse data does not guarantee competence on any individual temporal primitive.

No unit tests: There is no existing benchmark analogous to software unit tests that verifies whether a model handles basic temporal building blocks correctly.

Invisible covariate gaps: Existing benchmarks rarely test whether models exploit available exogenous covariates (leading indicators), so systematic failure to use causal information goes undetected.

Core Mechanism

SimpleTimeBench comprises three test families:

- *Monotonic trend:* Univariate series $x_t = \alpha t + \sigma \epsilon_t$ with varying slope α and noise σ . The oracle continues the trend; reversals or early flattening constitute failures.
- *Periodic signals:* Series with known periods (daily/weekly/annual) and controlled amplitudes. A model with temporal periodicity inductive bias should closely track cycles.
- *Leading indicators:* Multivariate series where channel x_{t-k} Granger-causes y_t with a known lag k , providing complete information for an accurate forecast.

Each family spans multiple noise levels, slopes, periods, and lag lengths to characterize sensitivity rather than produce a binary pass/fail verdict.

Technical Formulation

For the monotonic trend test at noise level σ and slope α :

$$x_t = \alpha t + \sigma \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1)$$

The oracle forecast for horizon h is $\hat{x}_{T+h} = \alpha(T + h)$, achieving $\text{MASE} \approx 1$. A model *passes* if $\text{MASE}_{\text{model}}/\text{MASE}_{\text{oracle}} < \delta$ for threshold $\delta = 2.0$.

For the leading indicator test with lag k :

$$y_t = \phi(x_{t-k}) + \sigma \epsilon_t$$

The oracle MSE is σ^2 . The leading-indicator utilization rate ρ is:

$$\rho = 1 - \frac{\text{MSE}_{\text{model}} - \sigma^2}{\text{MSE}_{\text{no-cov}} - \sigma^2}$$

where $\text{MSE}_{\text{no-cov}}$ is the MSE of a univariate model that ignores x . A ρ near 1 indicates full utilization; TSFMs average $\rho < 0.4$.

Learning or Inference Procedure

SimpleTimeBench is evaluation-only. The procedure:

1. Generate controlled synthetic series at multiple difficulty levels.
2. Run each TSFM (Chronos-2, Moirai, Toto) in zero-shot mode with the model’s default context window.
3. Compute MASE, sMAPE, and ρ for each test family.
4. Compare against oracle forecasters and classical baselines (ARIMA, Theta, ARIMAX).
5. Validate on real-world sensor datasets containing the tested primitives.

What the Guarantee Says

SimpleTimeBench provides a falsifiability guarantee: failure on any test family certifies the existence of natural inputs (with no distributional shift) for which the model underperforms a trivially simple oracle. This is a constructive, worst-case characterization—not a claim about average performance.

Experimental Findings

- **Trend reversal:** Chronos-2 reverses an increasing trend in $\approx 18\%$ of test cases at low noise; Moirai flattens trends early in $\approx 23\%$.
- **Periodic signals:** All three models exhibit MASE ratios above 2.0 relative to the oracle in $> 30\%$ of annual-period instances.
- **Leading indicators:** Average utilization rate $\rho < 0.40$ for all three TSFMs; ARIMAX achieves $\rho > 0.80$ on the same series.
- **Real-world sensors:** On building energy meters, TSFMs consistently underperform a tuned Fourier decomposition model on strongly periodic signals.
- **Classical model comparison:** ARIMA and Theta match or exceed TSFMs on monotonic trend and leading indicator tests despite being far simpler.

Ablations and Interpretation

Context window: Longer context modestly improves periodic performance (5–10% MASE reduction) but does not eliminate failures.

Model size: Larger model variants do not substantially improve SimpleTimeBench pass rates—the blind spots are not simply a parameter-count issue.

Fine-tuning: A small amount of domain fine-tuning on

primitive-containing examples raises pass rates to $> 85\%$, confirming the biases are learnable but absent in zero-shot pre-trained form.

Reference

Nafiseh Ghoroghchian, Haipeng Zhang, Shuyi Han, Alex Labach, George Stein. **Foundations without Fundamentals: Zero-Shot Blind Spots in Time Series FMs.** arXiv:2610.02058, October 2026.
<https://arxiv.org/abs/2610.02058>